

Xinyue YU (Christina)

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Curriculum Vitae
Updated in August 2026

EDUCATION

Purdue University, Edwardson School of Industrial Engineering <i>Ph.D., Ross Fellowship Awardee, advised by Dr. Nan Kong and Dr. Yuehwern Yih</i>	West Lafayette, IN, USA <i>Sept. 2025 – May 2030 (expected)</i>
The University of Hong Kong <i>Bachelor of Science (Major in Mathematics), Lee Shau Kee Scholar, First Class Honors</i>	Hong Kong S.A.R. <i>Sept. 2021 – June 2025</i>
McGill University <i>Exchange Student</i>	Montreal, Canada <i>Jan. 2023 – Apr. 2023</i>

RESEARCH INTERESTS

Operations Research, Resource Allocation, Bilevel Mixed Integer Programming, Algorithm Design

RESEARCH PROJECT

A Single-Leader Multi-Follower BMIP Approach to Government Incentive Optimization

Supervised by Dr. Nan Kong and Dr. Yuehwern Yih, Purdue University *Sept. 2025 – Present*

- Assisted in formulating the optimal subsidy allocation problem with a single-leader–multiple-follower bilevel integer programming model for the resultant Stackelberg game, where the government acts as the leader and each health system responds optimally and independently as multiple followers serving mutually disjoint areas, assisted in designing a decomposition-based algorithm that decomposes the formulated bilevel integer program into single-leader–single-follower subproblems and conducting instance testing

Optimal Control for Multi-Agent System (Final Year Project Research)

Supervised by Dr. Tak Kwong Wong, The University of Hong Kong *Aug. 2024 – May 2025*

- Referred to literature in linear quadratic discrete choice problems, explored further on realistic applications in optimal exits design for theaters and auditoriums
- Studied in the collective choice model (CCM) introduced in literature of linear quadratic mean field game (LQ-MFG), compared its strengths and drawbacks with other formulations

AI Prototype for Mental and Resources Support for Care Givers in Hong Kong

Worked with Agewhale, a startup in care givers support in Hong Kong *Aug. 2024 – May 2025*

- Cooperated with Agewhale to develop technical solutions for family carers to navigate and coordinate elder care approaches and resources in Hong Kong, offered accurate and updated information on related inquiries by the AI model
- Supported in data collection, sorting, and filtering, formulated the customer journey of users and the database, optimized the answers provided by the AI model by clearing redundant data and rephrasing wordings

Supervised Learning in Low Light Signal Detection Biomedical Experiments

Supervised by Dr. Vyacheslav Kalchenko, Weizmann Institute of Science (Israel) *Aug. 2022*

- Cowrote the report *Making the Invisible Visible: Regression and Artificial Neural Networks in Low Light Signal Detection*
- Diagnosed the overuse of live animals in low-light detection biomedical experiments and developed ML approaches
- Applied supervised learning to train a model on the imitation of human visual recognition and animal reactions

TALKS

A Single-Leader Multi-Follower Bilevel Programming Approach to Government Incentive Optimization

International Conference on Bilevel Optimization (ICBO) 2026, Pittsburgh, USA *August 2026*

TEACHING

Teaching Assistant, The University of Hong Kong

MATH2822 Mathematical Methods for Actuarial Science II	<i>Jan. 2025 – May 2025</i>
MATH1821 Mathematical Methods for Actuarial Science I	<i>Sept. 2024 – Dec. 2024</i>
COMP1117 Computer Programming (Python)	<i>Sept. 2023 – Dec. 2023</i>

HONORS, AWARDS & MEMBERSHIP

Purdue University

Ross Fellowship, assistantship for doctoral students Sept. 2025 – July 2029

The University of Hong Kong

Lee Shau Kee Scholarship, full tuition scholarship awarded by HKU & Lee Shau Kee Foundation Sept. 2021 – July 2025

Rosita King Ho Scholarship, exchange scholarship awarded by HKU & Rosita King Ho Foundation May 2023

Dean's Honor List, awarded by HKU Faculty of Science Sept. 2021 – May 2022

Young Scientist Scheme Member in HKU Faculty of Science Sept. 2021 – May 2023

Croucher Foundation

Full sponsorship in the International Science Summer Institute hosted in Weizmann Institute of Science (Israel) July 2022

Chinese Mathematical Society

Provincial First Prize in China National High School Mathematics Olympiad Oct. 2020

WORK EXPERIENCE

China International Capital Corporation

Equity Research Intern Shanghai, China
May 2024 – Aug. 2024

Petrel Capital Greater China Fund

Equity Research Intern Hong Kong S.A.R.
June 2023 – Aug. 2023

Yunnan Baiyao Group Pharmaceutical E-Commerce Co., Ltd.

Data Lake Project Intern Kunming, China
May 2022 – July 2022

LEADERSHIP, ACTIVITIES & SERVICES

INFORMS Student Chapter at Purdue University

Social and Outreach Officer West Lafayette, USA
Mar. 2026 – Present

The University of Hong Kong and Faculty of Science

Student Ambassador Hong Kong S.A.R.
Sept. 2021 – Present

The University Business Consulting Club

Vice President & Leader of Training Department Hong Kong S.A.R.
Apr. 2023 – Apr. 2024

PwC's Metaverse Challenge 2023

Team Member of Audience Favorite Hong Kong S.A.R.
July 2023

Sunflower Program, Shenzhen Shiyu Children Foundation

New Media Group Member and Social Work Volunteer (140 hrs in Kunming, 20 hrs in Shenzhen) Kunming & Shenzhen, China
July 2018 – Present

SKILLS & CERTIFICATES

Languages: English, Mandarin (native), Cantonese (basic)

Interests: Art curating, crochet, movies and TV series, music (R&B, popular), calligraphy, violin

Certificates: CFA ESG Investing Certificate, Society of Actuaries P Exam Passed